Dewen Zeng

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Research Interests

- Data miming and embedded computing

Education

Huazhong University of Science & Technology, Wuhan, China 2016 - present
Advisor: Prof. Yu Hu

- M.S. in Microelectronic Engineering
- GPA: 88.7/100

Huazhong University of Science & Technology, Wuhan, China 2012 - 2016

- B.S. in Optoelectronic Information Engineering
- GPA: 84.7/100

Research Experience

Distance accelerator for Time Series Mining, HUST 2016 - 2017
Research Student **Advisor: Prof. Yu Hu**

- Proposed a high-throughput reconfigurable memristor-based distance accelerator (MDA) for several distance functions in data miming (including dynamic time warping, longest common subsequence, etc.)
- Extracted the basic primitives to facilitate various distance functions to save chip area and optimized the analog circuit design for each specific algorithm. Compared to existing works, the MDA shows an overall speedup of 3.5x-376x on performance and an improvement of 1-3 orders of magnitude on energy efficiency.

Publications

Xu, X, Lin, F, Xu, W, Yao, X, Shi, Y, **Zeng, D**, & Hu, Y (2018). MDA: A Reconfigurable Memristor-based Distance Accelerator for Time Series Mining on Data Centers. IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems. (corresponding author)

Xu X, **Zeng D**, Xu W, et al. An efficient memristor-based distance accelerator for time series data mining on data centers[C]//Design Automation Conference (DAC), 2017 54th ACM/EDAC/IEEE. IEEE, 2017: 1-6.

Internship

Wuhan Easylinkin Technology Co., Ltd. Jan.2017 – Sep.2017

- Worked on a low-power communication protocol called Ibeacon designed by Apple and investigated an efficient algorithm for Indoor locating solution. Optimized the locating accuracy to around five meters.
- Developed a management system by using Java for sorting and computing real-time data collected from IoT devices. Worked on a demonstration system to simulate the real world company environment for locating and managing

employees' attendance and performance.

Course Projects

FPGA-based simple Digital Oscilloscope Mar.2015 – Apr.2015

- Designed a circuit system integrated a sample-hold circuit, a controllable gain amplifying circuit and a trigger circuit, which can detect sine wave, square wave and triangle wave with the voltage magnitude range from 8mV to 8V.
- Implemented a controlling system based on FPGA DE0-nano designed by Altera for generating clock and controlling ADC. Demonstrated the waveform on a TFT screen controlled by Nios II processor on the FPGA.

STM32-based Pulse Oximeter Aug.2015 – Sep.2015

- Implemented a detection circuit to collect pulse signal from the pulse detection device and a multi-stage amplifying and filtering circuit system for amplifying minor pulse signal and eliminate irrelevant signal.
- Investigated a low pass filter based on FFT on a stm32 platform and implemented an algorithm for computing the pulse rate and oxygen saturation. Obtained relative error of around 15% and 20% for pulse rate and oxygen saturation respectively compared to a sophisticated pulse oximeter device.

Awards

- Second prize, National Undergraduate Electronics Design Contest(NUEDC), 2015
- Outstanding undergraduate, Qiming College, Huazhong University of Science and Technology, 2016

Computer Skills

Programming: C/C++, Java, Javascript, Verilog, Matlab
Software: Intellij, Myeclipse, Visual Studio, Quartus ii, Keil